

Multistage PCA Whitening: A Robust Method to Dimensionality Reduction in Image Retrieval

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Abstract—Generating compact and robust feature representations using principal component analysis (PCA) is crucial for image retrieval tasks. However, most existing methods require PCA parameters to be learned from auxiliary datasets, which inevitably increases computational cost and limits generalization. To address this issue, we propose a novel dimensionality reduction learning method, namely multistage PCA whitening (MSPW), for image retrieval. Three main highlights are: 1) we propose a feature self-learning (FSL) method to learn PCA whitening (PW) parameters. This method can reconstruct the features of the retrieval dataset via singular value decomposition (SVD) and noise perturbation, which eliminates dependence on auxiliary datasets and alleviates performance degradation in high-dimensional features; 2) unlike existing single-stage dimensionality reduction methods, we introduce an online query self-learning (QSL) method that dynamically learns PCA parameters by incorporating query features, significantly improving the retrieval performance of using short-vector features; and 3) we propose a feature fusion (FF) method via using dimensional weighting to balance the contributions of various heterogeneous features, thereby enhancing the robustness of features across different dimensions. Experimental results on six benchmark datasets demonstrated that our MSPW method significantly outperforms existing state-of-the-art methods used for dimensionality reduction. Notably, our MSPW method using 4-D features can achieve more than 10% relative improvement over the previous best methods in terms of mean average precision (mAP) on two large-scale datasets.

Index Terms—Dimensionality reduction, feature fusion, image retrieval, principal component analysis whitening, singular value decomposition.

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I. INTRODUCTION

IMAGE retrieval focuses on searching for similar or identical images from a large-scale database given a query. As the number of images grows, achieving accurate and efficient retrieval becomes increasingly challenging. Early content-based image retrieval (CBIR) methods employed global features (e.g., color [1], texture [2], and shape [3]), as well as the integration of multiple global descriptors [4], [5] to represent image content. However, global features are easily affected by image rotation, scaling, and illumination. To overcome these limitations, researchers introduced the scale-invariant feature transform (SIFT) [6] and the Bag-of-Words (BoW) [7] models for instance-level image retrieval [8], where local features and their structural information are extracted and then quantized into fixed-length vectors, which improved the performance of CBIR. Unfortunately, they require a huge computational burden and are unsuitable for retrieval tasks on large-scale datasets.

Deep learning techniques have recently achieved remarkable success in various tasks [9], [10]. Nowadays, an increasing number of researchers are employing deep models for CBIR [8], [11]. The predominant strategy is to leverage pretrained [12] or fine-tuned [13] models to yield representations, followed by comparing them using the L2 distance metric. Unlike traditional handcrafted global and local features, this strategy captures higher-level abstractions and semantic information through end-to-end learning, thereby providing a more powerful representation for CBIR. However, most state-of-the-art methods primarily focus on enhancing feature discriminability through fine-tuning or retraining a model [14], [15], [16], often ignoring the impact of dimensionality reduction on retrieval performance—especially when compact, low-dimensional descriptors are required in large-scale scenarios.

Recent studies [17], [18], [19], [20] have explored dimensionality reduction by learning PCA-whitening parameters. Estimating these parameters directly from the target dataset's high-dimensional features does not require additional regularization or data augmentation; however, performance can degrade as dimensionality increases [21]. In these methods, principal component analysis (PCA) parameters are learned from an auxiliary dataset that resembles the target dataset, and the learned mapping matrix is shared by both database and query features during dimensionality reduction (see Fig. 1). Although effective, this offline learning paradigm suffers from two fundamental limitations: 1) it relies on externally collected auxiliary data, which requires substantial human effort and is often impractical in diverse and complex visual scenarios—for example, collecting suitable auxiliary data is very difficult for fine-grained retrieval [22]; and 2) it applies a single, fixed

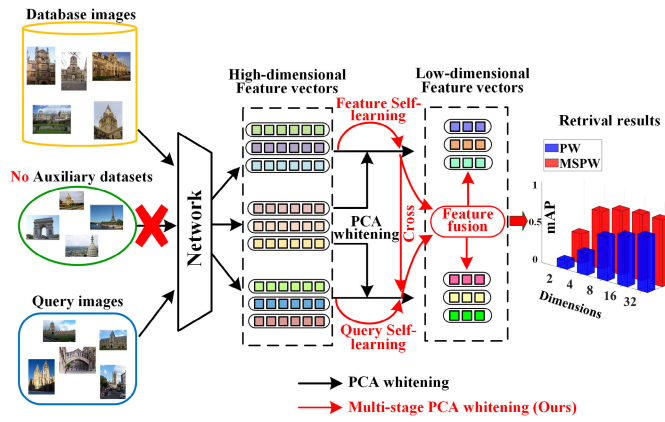


Fig. 1. Illustration of PW and the proposed MSPW. The MSPW method can eliminate the dependence of PW on the auxiliary dataset and significantly improve the retrieval performance of using low-dimensional features.

PCA transformation to all queries, ignoring the specific feature distribution of each query. Even within a single retrieval framework, complementary information is often distributed between features learned from the database and those from the queries, and fusing such information into a compact yet discriminative representation is nontrivial.

These observations motivate us to design a dimensionality-reduction framework that: 1) learns PCA-whitening parameters without any auxiliary dataset; 2) adapts the transformation to each query's feature distribution; and 3) supports robust fusion of heterogeneous features in a compact form. To this end, we propose a generalized multistage PCA-whitening (MSPW) method for dimensionality reduction. First, we propose a feature self-learning (FSL) module that uses singular value decomposition (SVD) and light noise perturbation to estimate PCA-whitening parameters directly from the retrieval dataset, eliminating reliance on external data and stabilizing estimation under high dimensionality. Second, we design a query self-learning (QSL) strategy that incorporates the query features into the PCA parameter learning process so that the transform adapts to the query's distribution and better preserves task-relevant information in low-dimensional spaces. Finally, to effectively combine retrieval-set features with query features, we introduce a dimensionality-weighted feature fusion (FF) scheme that dynamically assigns weights to multiple heterogeneous features, producing compact and discriminative descriptors.

To the best of our knowledge, this is the first work that jointly exploits database (retrieval) features and query features during the learning of dimensionality-reduction transforms for image retrieval. In summary, the main contributions are as follows.

- 1) We propose an FSL method that learns PCA-whitening parameters from the retrieval dataset itself via SVD and noise perturbation, thus removing the need for auxiliary data and alleviating performance degradation caused by high-dimensional features.
- 2) We introduce a QSL method that incorporates query features into the PCA parameter learning process, dynamically adjusting the transformation to better preserve important information in low-dimensional spaces and significantly improving short-vector retrieval performance on large-scale datasets.

- 3) We further propose an FF method that performs dimensionality-weighted fusion of heterogeneous features from the database and queries, producing compact, robust, and discriminative representations by effectively balancing the strengths of different features.

The remainder of this article is organized as follows. Section II reviews related work. A detailed description of the proposed methodology is provided in Section III. Experimental results, including performance evaluation and comparisons, are presented in Section IV. Finally, the conclusions are summarized in Section V.

II. RELATED WORKS

In this section, we first introduce the image retrieval methods based on deep features and then outline some representative methods of dimensionality reduction in CBIR.

A. Deep Feature Aggregation Methods

Earlier studies primarily relied on hand-crafted features, as documented in the literature [8]. Recently, the rapid advancement of deep learning techniques has significantly improved the performance of CBIR. Deep networks are commonly employed to extract deep features, which can be categorized into two groups: pretrained models and fine-tuned models. For example, Kalantidis et al. [23] proposed a representation method by applying spatial and channel weighting to emphasize high responsive regions and modulate the impact of channel bursts. Similarly, Xu et al. [24] selected the discriminative feature maps to enhance the target object features via suppressing background noise. Additionally, several recent methods have achieved more discriminative representations through multi-FF. For instance, Ma et al. [25] employed cross-entropy normalization to enhance the effectiveness of multi-FF, while Liu et al. [12] addressed the negative impact of texture bias by integrating shape, color, and deep semantic features. Hu and Bors [16] introduced a coattention mechanism to reweight the feature maps, enabling more accurate localization of the query regions. The MMFID method [26] introduced a modular multifeature integration framework that effectively combines global features, deep features, and semantic information, making full use of the advantages of different feature types. Similarly, Taheri et al. [27] considered multiple feature aspects and utilized the neutrosophic decomposition and deep map estimation to fuse features at different levels, resulting in more precise detailed information.

B. End-to-End Fine-Tuning Methods

Recent methods have demonstrated that the parameters of fine-tuned models on the specific datasets can significantly improve retrieval performance. Radenović et al. [28] trained CNN models by mining the positive and negative examples, which enhanced object-specific perception. Min et al. [29] proposed a two-stage triple-network training framework to enhance the learning capability of CNN models through the application of sorting loss and classification loss. To improve compatibility and discrimination between legacy and novel models, Pan et al. [30] employed adversarial learning along with flexible boundary constraints to mitigate distributional discrepancies across different models. Agarwal et al. [31] combined the advantages of supervised and self-supervised

learning by employing a hybrid learning method to train the network. To further improve the performance of initial retrieval rounds, visual reranking techniques are employed. Ouyang et al. [32] proposed a self-concerned reranking model that mines contextual relationships within ranked lists, thereby improving the model's robustness through feature enhancement. Tan et al. [33] introduced a reranking morphing network that can be jointly optimized with a feature extractor to improve the distinguishability of features. Additionally, Shao et al. [34] integrated global features into the reordering process and refined these global features, thereby significantly enhancing computational efficiency and storage utilization.

C. Feature Dimensionality Reduction Methods

Dimensionality reduction techniques can yield more compact representations and make them widely applicable to image retrieval. Jégou and Chum [20] employed PCA whitening to eliminate correlation and cooccurrence of visual word vectors. Wei et al. [22] used SVD whitening instead of PCA whitening (PW), effectively compressing feature dimensions and maintaining retrieval performance. To balance the variance along the direction of residual vectors and enhance the separability of global descriptors, Husain and Bober [35] employed clustered PCA and whitening to eliminate correlations among the residual vectors.

To facilitate the use of the PCA algorithm to process the high-dimensional sparse data, Ma and Yuan [36] optimized the PCA algorithm by filtering data through information entropy, thereby eliminating noninformative features. Radenović et al. [28] further improved the performance of fine-tuning methods by learning whitening parameters directly from the training data. To bypass feature post-processing, Cao et al. [37] learned low-dimensional local descriptors through convolutional self-encoding modules. Kalantidis et al. [38] adapted a self-supervised learning framework for the dimensionality reduction, which combines it with domain embedding learning to improve linear dimensionality reduction.

Leticio et al. [39] measured the similarity between images using data projection and visualization methods and then integrated various features using a postfusion method. Using PW for multifeature learning has emerged as a significant trend. For instance, hybrid PCA-whitening using multiparameter learning Hybrid PW (HPW) [17] and twice PCA whitening (TPW) [18] leveraging both semantic similarity datasets and retrieval datasets to learn PCA parameters, thereby addressing the degradation problems of performance associated with dimensionality reduction. Conversely, some methods (e.g., PCA-p-Whitening (PCApw) [40] and Gain Whitening Learning (GWL) [12]) focus on directly learning PCA parameters on the retrieval datasets, which aim to address the dependence on specific dataset.

The existing dimensionality reduction methods often rely on auxiliary datasets to learn the parameters of PCAs. Moreover, most methods focus on the retrieval of the dataset itself and are unable to effectively address the problem of performance degradation caused by high-dimensional features. Consequently, three critical issues need to be urgently addressed.

- 1) Eliminating the reliance on auxiliary datasets to improve the generalizability of dimensionality reduction methods.

- 2) Effectively mitigating the performance degradation of using high-dimensional features.
- 3) Improving the retrieval performance of short vector features on large datasets. Therefore, this study will focus on the above three aspects.

III. PROPOSED METHOD

The objective of this study is to propose a dimensionality reduction method that operates independently of any additional datasets and provides short vector features with robust discriminatory capability. Furthermore, this method acts as a type of feature postprocessing technique; it can accept inputs from any neural network and be seamlessly integrated with various CBIR methods.

The proposed MSPW method is shown in Fig. 2. It comprises a hierarchically progressive three-stage architecture, including FSL, QSL, and FF. These stages correspond to feature modeling, query enhancement, and multisource FF, respectively.

Stage I: FSL executes offline. Gaussian noise is introduced into database features and then they are reconstructed using SVD, yielding representations with richer structural information. Subsequently, these reconstructed features are decorrelated and standardized via PW to enhance their discriminative capability.

Stage II: QSL operates online in retrieval. SVD is applied to extract the latent information in query features and integrating it with the offline reconstructed features (FSL) to enhance their discriminative power and adaptability in low-dimensional space.

Stage III: FF aligns and integrates heterogeneous feature sets through dimension alignment and weighted fusion, and then achieves online optimization. This enables complementary integration of feature sets, thereby improving both the robustness of feature representation and the retrieval performance.

By structuring our proposed method into a hierarchically progressive three-stage framework, each stage has different strategies and execution times. We explicitly decompose the feature processing task into modular components. This design enables stage-specific optimization and coordination, resulting in a clearly defined and modular MSPW method. Sections III-A–III-C will elaborate on the implementation details and key designs of each stage.

A. Feature Self-Learning

Existing works have demonstrated that learning PCA parameters on auxiliary datasets with semantic similarity to the retrieval dataset can significantly improve retrieval performance. To eliminate reliance on auxiliary datasets, we propose a novel FSL method that learns PW parameters by directly generating semantically similar features on the retrieval dataset.

Let there be n images on a large-scale dataset. We utilize the DINOv2-reg [41] (ViT-L/14) model to extract deep features, which provides a set of high-dimensional vectors $X(n, D)$, where D represents vector dimensionality. The proposed FSL method consists of the following steps.

- 1) *Feature Reconstruction:* The feature vectors $X(n, D)$ are first normalized, followed by the addition of Gaussian noise to

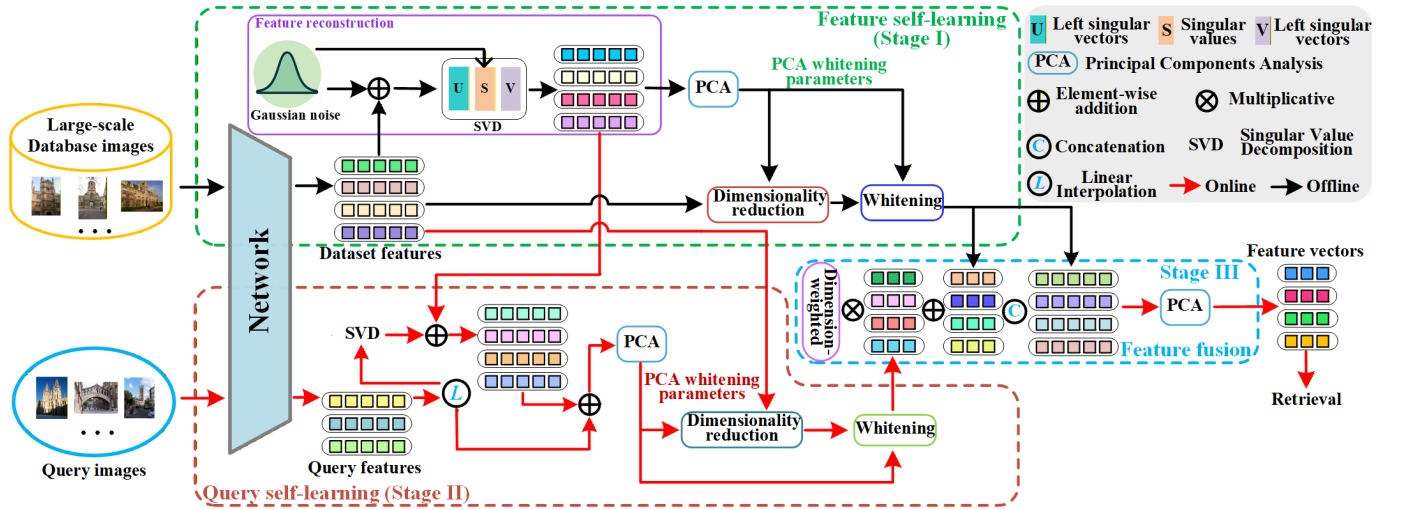


Fig. 2. Illustration of the proposed MSPW method.

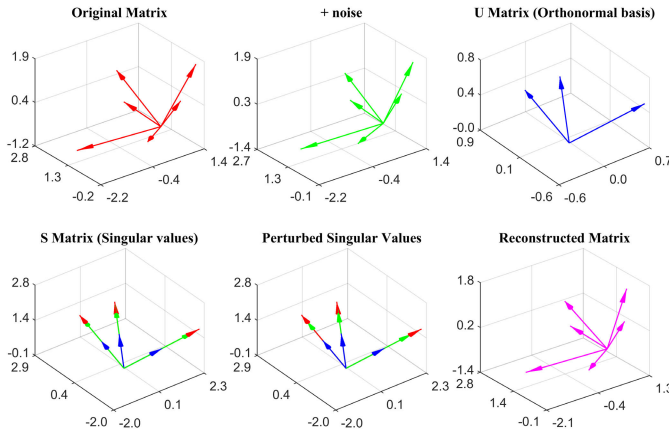


Fig. 3. Illustration of the feature reconstruction process in the proposed method. (using a 6×3 matrix as an illustrative example).

generate new auxiliary features. This process can be formally expressed as

$$X'(n, D) = \text{Norm}\{X(n, D)\} + \tau \mathcal{N}(0, 1) \quad (1)$$

where $\text{Norm}\{\cdot\}$ denotes L2 normalization, $\mathcal{N}(0, 1)$ denotes a Gaussian random variable with zero mean and unit variance, and τ denotes the noise level. In the above equation, the matrix $X'(n, D)$ can be decomposed using SVD. It is defined as follows:

$$X'(n, D) = USV^T \quad (2)$$

where U and V are the $n \times n$ and $D \times D$ orthogonal matrices, respectively. And S is a $n \times D$ diagonal matrix containing the singular values of $X'(n, D)$. To generate new feature representations, we perturb the singular values S as

$$S' = S + \tau \mathcal{N}(0, 1). \quad (3)$$

The perturbed matrix can be reconstructed as

$$X''(n, D) = US'V^T \quad (4)$$

where $X''(n, D)$ is the reconstructed features matrix with the perturbed singular values. To better illustrate the process of

feature reconstruction, we provide a visual representation. As shown in Fig. 3, the reconstructed features are generated by adding Gaussian noise and perturbing the singular values. This introduces controlled changes while preserving the original features. Consequently, this process produces a new feature matrix that retains semantic similarity to the original features.

2) *Principal Component Analysis*: We perform PCA on $X''(n, D)$, which is defined as follows:

$$P(n, d) = (X''(n, D) - \mu) \times V_f \quad (5)$$

where μ is the mean vector of $X''(n, D)$, and V_f is the eigenvector matrix of the covariance matrix of $X''(n, D)$, and d represents the number of retained principal components after dimensionality reduction. Subsequently, the covariance matrix of $P(n, d)$ can be computed as follows:

$$\sigma = \frac{1}{n-1} \sum_{i=1}^n (P_i - \bar{P})(P_i - \bar{P})^T \quad (6)$$

where $\bar{P}$ represents the mean vector of $P(n, d)$. SVD is then performed on σ to provide the left singular value matrix U_f and the diagonal matrix λ_f ; U_f and λ_f will be used in the subsequent whitening process. Finally, PCA is applied to $X(n, D)$ using μ and V_f , as defined below

$$X_{pca}(n, D) = (X(n, D) - \mu) \times V_f. \quad (7)$$

3) *Smooth Whitening*: To remove correlations between the features, we utilize whitening learning to process $X_{pca}(n, D)$. First, we perform a rotational transformation on $X_{pca}(n, D)$ by projecting it into the eigenvector directions, as follows:

$$X_{rot}(n, D) = X_{pca}(n, D) \times U_f. \quad (8)$$

The definition of $X_{rot}(n, D)$ is as follows:

$$X_{pw}(n, D) = \text{diag} \left(\frac{1}{\sqrt{\text{diag}(\lambda_f) + \varepsilon}} \right) \times X_{rot}^T(n, D) \quad (9)$$

where ε is a small constant used to maintain numerical stability, $\text{diag}(\cdot)$ denotes the whitening coefficient. In previous works [20], [21], ε is typically set to 10^{-5} , while here we set

$\varepsilon = 10^{-1}$, which controls the scaling of the diagonal elements of the matrix λ_f . This adjustment can disperse the distances between the feature vectors after whitening. At last, using (5)–(9) again, we can obtain the d -dimensional representation $X_{pw}(n, d)$, this can allow us to fully exploit the discriminative information across different vector dimensions.

B. Query Self-Learning

Existing works [16], [42] have demonstrated that extracting features from interest regions of queries can reduce interference from irrelevant information and further enhance retrieval performance. Inspired by this view, we propose a QSL method by incorporating query features into the PW process.

Let there be m query images, we first extract features from these images using DINOv2-reg, resulting in a set of query features $Q(m, D)$. To ensure that the size of the query features aligns with that of the database image features, we apply linear interpolation to expand $Q(m, D)$ to $Q(n, D)$. Next, we use (2)–(4) to reconstruct the query features $Q(n, D)$ to obtain features $Q'(n, D)$ and integrate the query features with $X''(n, D)$, as follows:

$$Q''(n, D) = \text{Norm}\{Q(n, D)\} + X''(n, D) + Q'(n, D) \quad (10)$$

we apply the same PW process to the FSL method. We conduct PCA on $Q''(n, D)$, which is defined as follows:

$$P'(n, d) = (Q''(n, D) - \mu_q) \times V_q \quad (11)$$

where μ_q is the mean vector of $Q''(n, D)$, V_q is the eigenvector matrix of the covariance matrix of $Q''(n, D)$, and d is the dimension of the features. The covariance matrix of $P'(n, d)$ is computed as follows:

$$\sigma' = \frac{1}{n-1} \sum_{i=1}^n (P'_i - \bar{P}') (P'_i - \bar{P}')^T \quad (12)$$

where $\bar{P}'$ represents the mean vector. SVD is then performed on σ' to provide the left singular value matrix U_q and the diagonal matrix λ_q . The μ_q and V_q are used to perform dimensionality reduction on $Q(n, D)$, as follows:

$$Q_{pca}(n, d) = (Q(n, D) - \mu_q) \times V_q. \quad (13)$$

We project $Q_{pca}(n, d)$ into the feature space spanned by using the eigenvectors U_q . This process is defined as follows:

$$Q_{rot}(n, d) = Q_{pca}(n, d) \times U_q. \quad (14)$$

Finally, we perform a whitening process on $Q_{rot}(n, d)$

$$Q_{pw}(n, d) = \text{diag} \left(\frac{1}{\sqrt{\text{diag}(\lambda_q) + \varepsilon}} \right) \times Q_{rot}^T(n, d) \quad (15)$$

where ε is 10^{-1} . The QSL method requires calculation through the online stage, the time taken for feature reduction is negligible compared to the time taken for retrieval using high-dimensional features. In addition, using short feature vectors for queries on large datasets has higher retrieval efficiency.

C. Feature Fusion

The database features X_{pw} and query features Q_{pw} each have advantages in different dimensional feature spaces. By combining them, we can improve the robustness across dimensional features. To achieve this goal, we propose an FF method that can dynamically integrate X_{pw} and Q_{pw} .

We observe that the features X_{pw} are more stable in high-dimensional spaces, while the query features Q_{pw} can significantly improve the retrieval performance in low-dimensional spaces (see Table I). Therefore, we introduce a dimension-weighting coefficient dw to dynamically adjust the contribution of Q_{pw} and X_{pw} , as follows:

$$F(n, d + D) = [dw \times Q_{pw}(n, d) + X_{pw}(n, d), X_{pw}(n, D)] \quad (16)$$

where $[\cdot]$ denotes the concatenation of feature matrices. The dimension-weighting coefficient dw is defined as follows:

$$dw = \frac{D}{1 + e^{d-\alpha}} \quad (17)$$

where α is a dimensionality control parameter. We observe that when learning the PCA parameters from the dataset itself, the performance peaks at a lower dimension (between 32 and 64) and starts to decrease (see Fig. 4), which is consistent with existing works [21], [40]. Therefore, we set $\alpha = 40$. When the dimensionality exceeds 40, the X_{pw} plays a decisive role. This process is conducted entirely during the offline phase and does not impact on the online query time. When the dimensionality is below 40, Q_{pw} becomes the dominant factor. Due to its low dimensionality, Q_{pw} does not consume too much online learning time (see Table II).

Since concatenating the feature matrices increases the vector dimension, we perform dimensionality reduction on F to provide a d -dimensional representation. It is defined as follows:

$$F'(n, d) = (\text{Norm}\{F(n, d + D)\} - \mu'') \times V'' \quad (18)$$

where μ'' and V'' are the mean vectors and the eigenvector matrices of the covariance matrix, respectively. Finally, normalizing $F'(n, d)$ yields a compact yet effective representation

$$F''(n, d) = \text{Norm}\{F'(n, d)\}. \quad (19)$$

Based on the proposed FF method, we can effectively integrate different whitening features and features of different dimensions to obtain a compact and highly discriminative short-vector feature representations. We will show the significant improvement brought about by our FF method in Table I.

IV. EXPERIMENTS

In this section, we conduct a hierarchical evaluation of the proposed MSPW method for image retrieval. Section IV-A presents the benchmark datasets, evaluation metrics, and implementation details. Section IV-B analyzes the core behavior of MSPW through hyperparameter sensitivity and overall performance across different pretrained models. Section IV-C provides ablation studies to quantify the contribution of each component. Section IV-D reports the complexity and efficiency analysis. Section IV-E compares MSPW with existing dimensionality reduction methods and state-of-the-art image retrieval approaches. Section IV-F further examines the generalization ability of MSPW in plug-and-play usage and fine-grained retrieval scenarios. Finally, Section IV-G offers a discussion of the empirical findings and limitations.

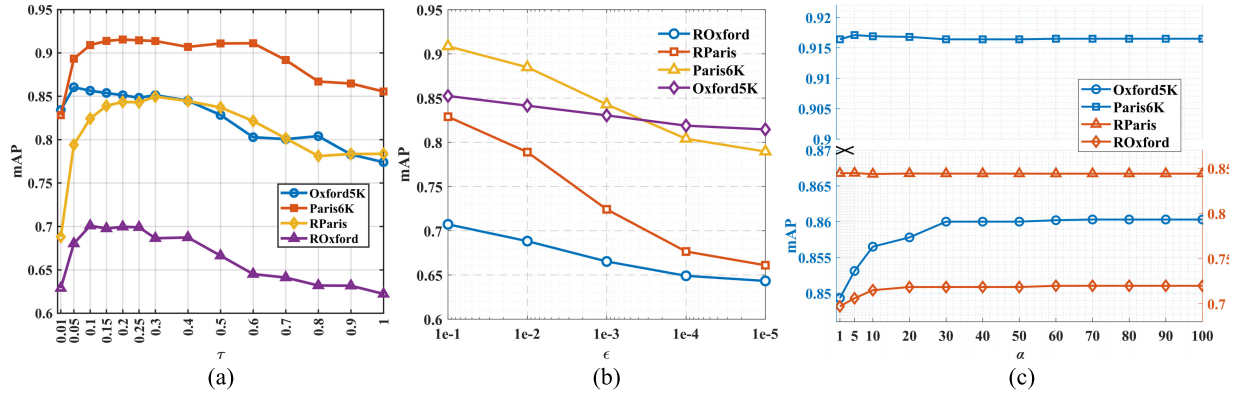


Fig. 4. Sensitivity analysis of hyperparameters τ , ϵ , and α on retrieval performance. (a) τ : noise level in feature reconstruction. (b) ϵ : small stability constant scaling the whitening coefficients. (c) α : dimensionality control parameter for the target feature dimension. We choose $\tau = 0.15$, $\epsilon = 0.10$, and $\alpha = 40$ as a good tradeoff between accuracy and robustness for subsequent experiments.

TABLE I
ABLATION ANALYSIS OF THE PROPOSED METHOD. THE BEST RESULTS ARE MARKED WITH BOLD

Component Settings	RParis (H)									
	2	4	8	16	32	64	128	256	512	1024
PW	10.56	27.68	55.15	63.00	69.83	79.34	82.33	83.29	83.92	85.25
Baseline (PWs)	30.27	61.81	74.14	81.54	76.60	74.40	68.83	58.29	45.29	31.22
+FSL	35.88	62.57	72.34	80.95	80.87	82.30	83.16	83.62	83.64	83.34
+QSL	36.54	71.33	79.53	81.47	81.03	81.45	80.73	80.95	81.14	80.43
+FSL+QSL	32.39	69.01	80.37	79.30	78.73	78.90	78.75	79.12	79.38	75.55
+FSL+QSL+FF	36.74	71.55	79.74	82.56	82.86	82.87	83.94	84.45	84.33	84.02
Component Settings	Paris6K									
	2	4	8	16	32	64	128	256	512	1024
PW	19.58	54.01	81.41	88.44	90.39	95.39	95.55	96.09	96.21	96.40
Baseline (PWs)	45.40	83.47	89.92	95.40	93.47	92.10	90.79	86.34	78.83	67.06
+FSL	49.04	84.17	89.37	95.46	95.61	95.81	95.96	96.07	96.13	96.17
+QSL	57.88	92.36	95.06	94.85	94.96	94.85	95.04	95.10	95.41	94.65
+FSL+QSL	55.38	90.08	93.45	95.24	95.22	94.95	94.93	95.25	95.21	95.23
+FSL+QSL+FF	57.83	92.39	95.06	95.42	95.62	95.68	96.02	96.14	96.21	96.22

TABLE II
COMPARISON OF THE COMPUTATIONAL EFFICIENCY OF DIFFERENT DIMENSIONALITY REDUCTION METHODS. THE SYMBOL ‘ $\uparrow$ ’ MEANS THAT THE ACCELERATED OPERATION IS PERFORMED USING THE RANDOM SVD [52]. DIMENSIONS = 4

Method	ROxford(H)				RParis(H)			
	Complexity	Time (s)	Memory (MB)	mAP (%)	Complexity	Time (s)	Memory (MB)	mAP (%)
PW	$O(nD^2+D^3)$	1.954	88.406	11.20	$O(nD^2+D^3)$	1.894	88.406	27.68
PWs	$O(nD^2+D^3)$	1.552	39.012	21.86	$O(nD^2+D^3)$	1.789	49.395	61.81
MSPW(Ours)	$O(nD^2+D^3)$	2.228/1.623 $\uparrow$	39.012	50.90/51.47 $\uparrow$	$O(nD^2+D^3)$	2.421/1.814 $\uparrow$	49.395	71.55/72.59 $\uparrow$

A. Experimental Setup

We first describe the benchmark datasets, evaluation metrics, and implementation details that are shared by all subsequent experiments.

1) *Datasets and Evaluation Metrics*: The **Oxford5K** dataset [43] consists of 5062 images of Oxford landmarks sourced from Flickr. It contains 11 categories, with five images per category, resulting in a total of 55 query images, each provided with a bounding box to indicate the region of interest. The images on this dataset are annotated with four labels: good, ok, junk, and bad. The **Paris6K** dataset [44] contains

6392 images of Paris landmarks. Like Oxford5K, 55 images from the Paris6K dataset are used as queries, representing 11 types of landmarks with five queries per type. Each query image is provided with a bounding box, and the images on this dataset are annotated with the same four labels as Oxford5K. To evaluate the robustness of our method, we also tested it on two large-scale datasets: **Oxford105K** and **Paris106K**. These datasets were created by augmenting the Oxford5K and Paris6K datasets with an additional 100 068 distracting images from the Flickr100K dataset, allowing for a more comprehensive evaluation of retrieval performance in a challenging, large-scale setting. The **ROxford** (ROxf) and

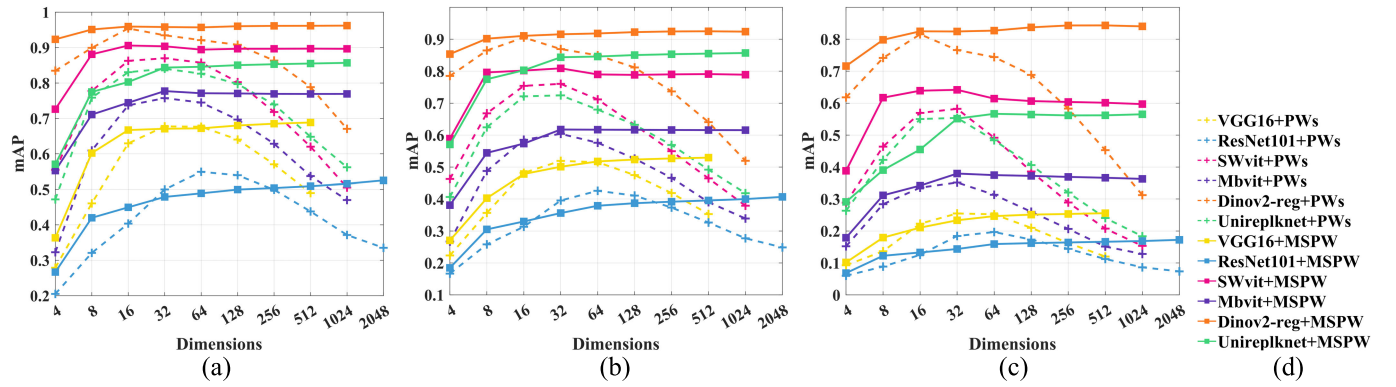


Fig. 5. Comparative performance of our MSPW method on different pretrained models. MSPW consistently improves retrieval performance over self-PW, especially at low feature dimensions.

RParis (RPar) datasets [45] were created by updating the annotations and evaluation protocols of the Oxford5K and Paris6K datasets, respectively, and adding 15 additional query images. According to [44], both ROxf and RParis are divided into three evaluation protocols: easy (E), medium (M), and hard (H).

2) *Implementation Details*: For PW, as in previous studies [18], [20], [23], we learned PW parameters from the Paris6K dataset and applied them to the Oxford5K and Oxford105K datasets. Similarly, parameters from Oxford5K were applied to the Paris6K and Paris106K datasets. The ROxf and RParis datasets are complementary to that of learning PCA parameters. PW self (PWs) [21] refers to learning PCA parameters directly from the dataset being evaluated. During the experiments, we used L2 distance for image matching, following prior works [8], [11], [13], and mean average precision (mAP) was adopted as the primary evaluation metric. For feature extraction, we primarily used the latest DINOv2-reg model in this study.

All algorithms are implemented in MATLAB R2022b and evaluated on a workstation equipped with an Intel¹ Xeon¹ Gold 6230 CPU @ 2.10 GHz and 24 GB RAM. No GPU acceleration is used, and all compared methods are executed under the same experimental environment.

B. Hyperparameter Sensitivity and Overall Performance

1) *Hyperparameter Sensitivity*: In Fig. 4(a), we observe that all four curves simultaneously peak at approximately $\tau \approx 0.15$; deviating from this value (either increasing or decreasing τ) results in a significant drop in overall performance. As shown in Fig. 4(b), $\varepsilon = 0.1$ consistently achieves the highest mAP scores on all four datasets. When ε is too small (<0.001), low-frequency noise amplification occurs, which causes rapid performance degradation. In Fig. 4(c), setting $\alpha = 40$ yields the highest mAP scores on all datasets. Through systematic parameter sensitivity experiments, we validated the rationality and generalization of the selected hyperparameters, specifically $\tau = 0.15$, $\varepsilon = 0.1$, and $\alpha = 40$.

2) *Overall Performance Across Different Pretrained Models*: We use various pretrained models to extract features, including VGG16 (pool5) [46], ResNet101 (res5c_relu) [47], Swin-L (norm) [48], MobileViTv2-200 (final_conv) [49], DINOv2-reg (norm) [41], and UniRepLkNet-L (norm) [50].

The specific layers used for feature extraction are indicated in parentheses for each model. To demonstrate the superiority of MSPW in dimensionality reduction, we perform a comparative experiment using PWs.

As can be seen from Fig. 5, the PWs method suffers a notable performance decline when using high-dimensional features. A possible explanation is that when learning PCA parameters from the test dataset itself, PCA may capture dataset-specific noise or irrelevant information. In contrast, our MSPW method not only retains the performance of high-dimensional features but also consistently enhances the retrieval accuracy of low-dimensional features. The MSPW method clearly outperforms PWs, demonstrating superior stability and effectiveness. Notably, when applied across different pretrained models, MSPW delivers more than a 10% improvement in mAP scores for the highest feature dimension, indicating its ability to significantly and consistently improve retrieval performance.

C. Ablation Studies

1) *Effectiveness of Different Components*: This section evaluates the effectiveness of the three key components of our proposed MSPW method through ablation experiments. These components are FSL, QSL, and FF. As shown in Table I, two baseline models (PW and PWs) were used as control experiments, with the proposed components incrementally added to the PWs baseline to assess their individual contributions.

We compress the feature vector to extremely low dimensions to evaluate the robustness of our method. First, the FSL component demonstrates a significant performance improvement when handling high-dimensional features ($\text{dim} > 64$). For example, using 1024-D features ($\text{dim} = 1024$) results in 52.12% and 29.11% improvements in mAP on the RParis (H) and Paris6K datasets, respectively. This suggests that the FSL component enhances the robustness of high-dimensional features through feature reconstruction. Second, the QSL component significantly improves the retrieval performance when working with low-dimensional features ($\text{dim} < 64$). For example, using only 2-D features ($\text{dim} = 2$), the QSL component improves mAP by 6.27% on RParis (H) and 12.48% on Paris6K. This improvement indicates that learning PCA parameters from query features helps retain critical query-relevant information, thereby increasing the discriminative power of low-dimensional features. As can be seen from performance comparison of FSL and QSL modules, FSL using high-dimensional features outperforms QSL, whereas

¹Registered trademark.

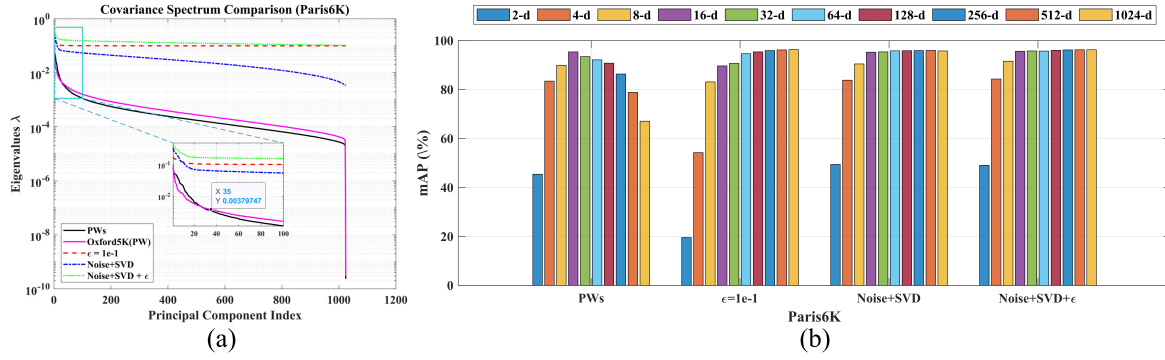


Fig. 6. Ablation analysis of FSL methods. (a) Covariance spectra on Paris6K under different whitening variants, showing how noise + SVD + ϵ prevent eigenvalues from collapsing. (b) mAP at different feature dimensions for PWs and the proposed variants on Paris6K, illustrating the recovery of high-dimensional performance.

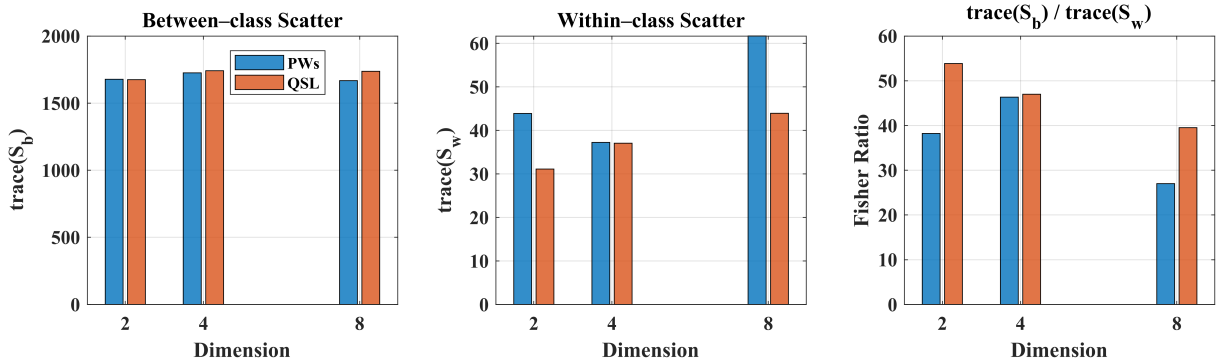


Fig. 7. Between-/within-class scatter and Fisher ratio of PWs versus QSL on the Paris6K dataset, showing that QSL yields lower within-class scatter and a higher Fisher ratio in low-dimensional spaces.

QSL using low-dimensional features outperforms FSL. Finally, the FF component combines the strengths of FSL and QSL, resulting in optimal retrieval performance in both high- and low-dimensional feature spaces. This highlights the ability of the FF component to further enhance the robustness of features at different dimensionalities. The results clearly indicate that using all the proposed components together yields the best overall performance.

2) *Ablation Analysis of FSL*: As shown in Fig. 6(a), when PW parameters are learned from the Oxford5K dataset itself, the eigenvalues of its high-dimensional features decay almost to zero, which indicates that the substantial noise exists in these high-dimensional features. According to (9), as $\lambda \rightarrow 0$, this noise is excessively amplified in the process of whitening, causing the noise components to dominate similarity computations and results in a drastic degradation of retrieval performance. Indeed, the proposed FSL method is a feature reconstruction method based on Gaussian noise injection and SVD reconstruction. When the value of ϵ is increased to smooth the eigenvalues and to avoid excessive amplification of noise, it thereby recovers the retrieval performance of high-dimensional features [as shown in Fig. 6(b)]. Fig. 6(a) further shows that auxiliary datasets like Oxford5K can provide effects similar to those of our generated features, as both boost high-dimensional eigenvalues without over-amplifying small eigenvalues, thus helping maintain the robustness of high-dimensional features.

3) *Analysis of QSL Effectiveness*: QSL integrates the variance of the query samples into the whitening transformation

by incorporating query features. Directions that exhibit high variance in the query samples correspond to larger eigenvalues λ_f , where samples of the same class are widely dispersed. Consequently, the whitening coefficient $1/(\text{diag}(\lambda_f) + \epsilon)^{1/2}$ becomes smaller, thereby automatically attenuating those directions and leading to a larger within-class scatter. As shown in Fig. 7, QSL outperforms PWs in low-dimensional space (e.g., 2, 4, and 8). The primary reason lies in the fact that it can simultaneously achieve a slight increase in inter-class separation and a significant improvement in intraclass compactness. QSL reduces intraclass variance by introducing noise perturbation and SVD reconstruction, and can effectively incorporate query features and enable the learned projection subspace to cluster similar samples. This significantly decreases intra-class variance and enhances the overall Fisher ratio, indicating improved discriminability of the projected features, thereby leading to better retrieval performance.

4) *Visual Analysis*: As shown in Fig. 8, we use t-SNE to visualize the distribution of deep feature representations in both the PW and MSPW methods. It is evident that the MSPW method reduces intra-cluster distances, increases inter-cluster distances, and enhances category separation. This indicates that our approach can more effectively distinguish between different image categories.

D. Comparison With State-of-the-Art Methods

As shown in Table II, the computational complexity of our proposed MSPW method is comparable to that of the PW and PWs methods, approximated as $O(nD^2 + D^3)$, where n

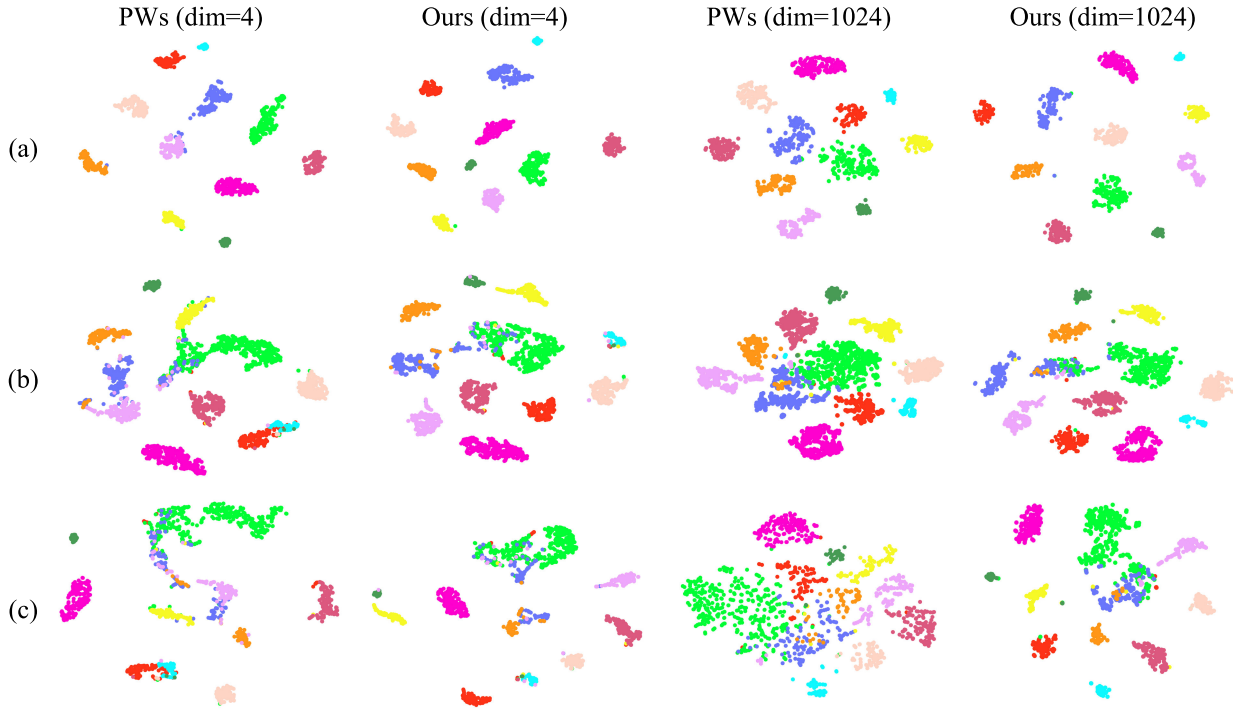


Fig. 8. T-SNE [51] on RParis: PWs versus MSPW at 4-D and 1024-D (columns) across easy/medium/hard queries (rows); MSPW yields tighter clusters and clearer interclass separation.

denotes the number of features and D represents the feature dimension. Although the MSPW method experiences a slight increase in runtime due to the additional PCA and SVD steps, this increase is effectively constrained to within 0.8 s when compared to the PWs method. When using the randomly structured SVD (rSVD) [52] for accelerated calculations, the MSPW method adds no more than 0.1 s, though this can introduce minor performance fluctuations. Furthermore, the memory footprint of the MSPW method is approximately half that of the PW method, as it avoids reliance on auxiliary datasets. Experimental results demonstrate that the MSPW method not only significantly improves retrieval performance but also maintains reasonable runtime and low memory consumption.

E. Comparison With State-of-the-Art Methods

1) *Dimensionality Reduction Baselines*: To evaluate the improvements achieved by the MSPW method, we conducted a comparison against seven state-of-the-art dimensionality reduction methods. These include four methods that rely on learning PCA parameters from auxiliary datasets: PW [20], HPW [17], Twice PCA Whitening (TPW) [18], and Adaptive PW (APW) [19]. Additionally, we considered a method trained within a self-supervised framework: Twin Learning for Dimensionality Reduction (TLDR) [38], as well as two methods that learn PCA parameters directly from the retrieval dataset: GWL [12] and PCApw [40].

As shown in Fig. 9, the MSPW method exhibits substantial performance improvements over existing approaches when using low-dimensional features (dimensions < 64). Specifically, on the large-scale datasets Oxford105K and Paris106K, our method achieves a mAP that exceeds the second-best method, TLDR, by 27.1% and 34%, respectively, at a feature

dimension of 2, without the need for additional training. For high-dimensional features, the performance of MSPW is comparable to methods that learn PCA parameters from auxiliary datasets. Both GWL and PCApw experience a significant performance drop when employing high-dimensional features on the ROxf and RParis datasets. By contrast, MSPW maintains stable retrieval performance across high-dimensional feature spaces and consistently outperforms both GWL and PCApw across all dimensions. This highlights the effectiveness of MSPW in mitigating the performance degradation commonly associated with high-dimensional feature representations.

2) *State-of-the-Art Image Retrieval Methods*: We compared our MSPW method with 11 state-of-the-art methods in different dimensions, including four feature aggregation methods based on pretrained models: MMFID [26], UTRL [18], DSCD [19], and TransVLAD [54], as well as seven fine-tuning methods: GeM [28], REMAP [55], AILIR [58], MABNet [30], Yang et al. [62], *siaSUM [64], and DUODIS-ws [65]. As shown in Table III, MSPW consistently outperforms all other methods across various dimensional settings. Specifically, our approach provides a 1% and 2.1% improvement in mAP on the Oxford105K and Paris106K datasets, respectively, compared to the second-best method, DUODIS-ws [65]. Notably, MSPW does not rely on models specifically designed to enhance representation capabilities, but rather focuses on providing a general postprocessing method.

Furthermore, we adapted the MSPW method to three representative methods (UTRL, DSCD, and PLRA) to conduct comparisons under identical network architectures. As can be seen from Fig. 10, MSPW consistently achieves significant performance gains. On the RParis dataset in terms of M protocol, MSPW improves mAP by + 9.6%, + 9.9%, and + 6.0% over UTRL, DSCD, and PLRA, respectively. These



Fig. 9. Performance comparison of different dimensionality reduction methods, the TLDR method uses the retrieval dataset to train the model. Experimental data for comparison methods generated from official codes. MSPW clearly outperforms existing methods in very low dimensions (<64) and maintains competitive or superior accuracy to training-based methods in high-dimensional settings.

results demonstrate that the improvements stem from the inherent mechanism of MSPW.

F. Cooperating With More Cutting-Edge Models

1) *Plug-and-Play on Cutting-Edge Models*: In order to demonstrate the generalization ability of MSPW, we combine MSPW with 16 advanced models, including 11 fine-tuned models focusing on image retrieval: DELG [37], [66], ViT-GaL [67], SOLAR [66], [68], DOLG [69], Token [66], CVNET [70], SuperGlobal [34], SENET [71], NetVLAD [66], [72], DUODIS-ws [65], and Whiten-MTD [73], along with five pre-trained models: CLIP [74], BLIP [75], IBOT [76], RADIOv2 [77], and DINOv2-reg [41]. As can be seen from Table IV, when the vector dimension is reduced by a factor of 8, our method can ensure that the retrieval

performance does not decrease by more than 1.4% for the fine-tuned models, and even slightly improves for the CVNET, SuperGlobal, SENET, and DUODIS methods. Furthermore, MSPW reduces retrieval time by half and requires only 1/4 of the original memory (average results of 5 runs). These results further validate the effectiveness and versatility of the MSPW approach in reducing vector dimensionality with minimal performance loss, while optimizing retrieval time.

2) *Extension to Fine-Grained Retrieval*: To further validate the generalizability of our MSPW method across different datasets, we conducted experiments on fine-grained image retrieval datasets. Unlike the tasks of instance-level image retrieval, which focuses on retrieving images containing the exact same object, fine-grained image retrieval targets images

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON FOUR BENCHMARK DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD.
[O]: OFF-THE-SHELF (PRETRAINED ON IMAGENET)

Method	Net	Train Set	Dim	Oxford5K	Paris6K	Oxford105K	Paris106K
MMFID [26]	VGG19	[O]	128	71.1	82.1	66.5	75.7
PLRA [17]	VGG16	[O]	128	77.9	86.2	66.0	75.6
UTRL [18]	RepVGG [53]	[O]	128	82.9	91.6	66.5	81.5
DSCD [19]	VGG16	[O]	128	82.2	89.6	72.5	81.3
GeM [28]	ResNeT101	SfM-120K [28]	128	79.5	84.5	72.2	74.3
REMAP [55]	ResNeXt101 [56]	Landmarks-retrieval [57]	128	81.1	-	79.3	-
MSPW(Ours)	DINOv2-reg (ViT-L/14)	LVD-142M [41]	128	92.45	96.03	87.8	94.03
MMFID [26]	VGG19	[O]	256	77.6	84.6	73.6	78.3
PLRA [17]	VGG16	[O]	256	78.3	86.0	71.4	77.9
UTRL [18]	RepVGG	[O]	256	83.9	91.0	76.2	85.5
DSCD [19]	VGG16	[O]	256	82.1	88.7	77.8	83.0
GeM [28]	ResNeT101	SfM-120K [28]	256	83.1	87.5	77.3	78.8
MSPW(Ours)	DINOv2-reg (ViT-L/14)	LVD-142M [41]	256	92.6	96.14	89.01	94.52
MMFID [26]	VGG19	[O]	512	77.3	86.1	74.1	80.0
PLRA [17]	VGG16	[O]	512	78.1	85.9	73.0	78.0
UTRL [18]	RepVGG	[O]	512	83.5	90.5	79.8	86.3
DSCD [19]	VGG16	[O]	512	81.7	87.9	78.9	82.0
TransVLAD [54]	VGG16	[O]	4096	76.40	82.30	-	-
GeM [28]	ResNeT101	SfM-120K [28]	512	84.6	90.0	80.4	82.6
AILIR [58]	VGG16	Landmark dataset [59]	2048	79.4	87.38	79.2	86.6
MABNet [31]	ViT-small [60]	GLDv2-train-clean [61]	-	91.5	94.2	88.7	90.4
Yang et.al [62]	ResNeT50	Landmarks-clean [63]	4096	90.6	90.1	88.2	87.7
-siaSUM [64]	VGG16	SfM-120K [28]	4096	84.4	89.3	82.3	85.0
DUODIS-ws [65]	VGG16	SfM-120K [28]	2048	90.00	95.2	89.0	92.4
MSPW(Ours)	DINOv2-reg (ViT-L/14)	LVD-142M [41]	512	92.75	96.20	90.0	94.52

from the same category [22], such as cars of the same brand but in different colors. Due to the substantial variations among fine-grained datasets, it is not feasible to provide similar auxiliary datasets for learning PCA parameters. Our proposed method overcomes the limitations of traditional PW, so it can be directly applied to fine-grained retrieval tasks.

3) *Experimental Setup*: We evaluate the performance of our method on three widely used fine-grained image retrieval datasets: CUB200-2011 [78], Stanford Dogs [79], and Stanford Cars [80]. The CUB200-2011 dataset contains 5794 images across 200 categories for testing. Similarly, the Stanford Dogs dataset contains 8580 images across 120 categories, while the Stanford Cars dataset contains 8041 images across 196 categories for testing. Features were extracted using the pretrained DINOv2-reg (ViT-L/14) model, where bilinear interpolation is applied to resize the input images to 448×448 pixels. Since the queries in the fine-grained retrieval datasets are identical to the test images, the QSL component was omitted during the execution of the MSPW method.

4) *Comparison and Analysis of Experimental Results*: We compared our MSPW method with seven other advanced feature reduction methods on three datasets, including PWs [21], SVD whitening [22], GWL [12], PCApw [40], uniform manifold approximation and projection (UMAP) [81], Minimum Distortion Embedding (MDE) [82], and TriMap [83], where the SVD whitening method was implemented by using our code. The comparison results are shown in Table V. It

is clear that the MSPW method can achieve the best retrieval performance on most fine-grained datasets. The runtime of the MSPW method is 17.88 s, it is slightly higher than that of traditional PWs (8.61 s) and SVD whitening (9.35 s), and it is 1.4–14 times faster than nonlinear methods such as GWL, UMAP, and TriMap. In terms of memory usage, MSPW occupies the same order of magnitude as most compared methods. It is only marginally higher than the lightest approach, PWs, and significantly lower than UMAP, which requires complex graph embedding. These results highlight the effectiveness and adaptability of the MSPW method in different scenarios.

G. Discussion

1) *Theoretical and Practical Value*: The proposed MSPW method combines theoretical innovation with practical benefits. In theory, the FSL module uses noise perturbation and SVD reconstruction to robustly estimate PCA parameters without any auxiliary dataset, breaking the long-standing reliance of conventional methods on external data. The QSL module further enables dynamic adaptation of the dimensionality reduction process to each query feature, thereby extending the application scope of PW. In practice, MSPW directly learns the statistics of the target data, which improves model stability, eliminates the need for external data and simplifies the deployment pipeline, while its compact representations reduce storage overhead and suit resource-constrained environments. Potential application scenarios for our

TABLE IV
COOPERATING MSPW WITH MULTIPLE SOTA MODELS. BOLD NUMBERS INDICATE A PERFORMANCE INCREASE.
†: OFFICIAL CODE. ◊: NONCROPPED QUERIES

Method	Pub.	Dim	EASY				MEDIUM				HARD				Mean	Diff	Time (s)	Memory (MB)
			ROxf		RPar		ROxf		RPar		ROxf		RPar					
			mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10				
Fine-tuning model																		
DELG [†] [28][66]	ECCV20	2048	91.89	93.16	95.00	96.08	78.55	89.89	88.58	97.57	60.89	74.73	76.05	91.43	86.15	-0.13	0.5	48.79
DELG [†] + MSPW	-	256	91.40	92.72	94.44	95.51	79.01	89.50	88.64	97.00	61.82	75.20	76.39	90.57	86.02		0.23	10.06
ViTGal ^{◊†} [67]	ACCV22	384	95.70	95.08	96.76	97.71	84.42	92.57	92.53	98.86	65.89	77.14	83.60	97.00	89.77	-1.33	0.25	12.83
ViTGal ^{◊†} + MSPW	-	48	94.65	93.03	95.73	96.45	82.99	91.14	91.84	98.29	64.78	75.86	82.41	94.14	88.44		0.19	5.12
SOLAR [†] [66][68]	ECCV20	2048	93.50	94.17	95.70	96.37	79.65	91.07	88.63	98.14	59.99	74.93	76.15	93.71	86.83	-0.55	0.49	48.96
SOLAR [†] + MSPW	-	256	93.22	93.66	95.23	95.80	79.83	90.86	88.34	97.14	60.20	74.43	75.82	91.86	86.28		0.23	10.1
DOLG [†] [69]	ICCV21	512	94.66	95.29	96.86	97.86	82.38	93.00	90.94	98.57	62.92	76.86	80.48	95.71	88.79	-0.54	0.26	15.59
DOLG [†] + MSPW	-	64	94.54	94.12	96.54	97.16	82.94	92.71	90.31	97.86	63.21	76.00	79.85	93.71	88.25		0.21	5.92
Token [†] [66]	AAAI22	1024	95.15	95.72	95.65	96.36	82.15	92.00	89.40	98.14	65.74	77.84	78.44	95.00	88.47	-0.21	0.34	26.65
Token [†] + MSPW	-	128	94.89	94.87	95.69	96.37	82.29	92.00	89.49	98.00	66.00	76.56	78.78	94.14	88.26		0.22	7.30
CVNET [†] [70]	CVPR22	2048	94.29	93.44	93.87	95.86	80.98	90.86	88.83	98.00	62.12	77.71	76.49	94.00	87.20	+0.34	0.51	48.74
CVNET [†] + MSPW	-	256	94.78	94.64	93.96	95.57	81.89	92.71	88.74	97.14	63.28	78.14	76.32	93.29	87.54		0.23	10.06
SuperGlobal [†] [34]	ICCV23	2048	94.66	93.65	93.98	95.29	83.68	92.27	90.36	96.57	67.56	79.68	80.33	94.29	88.53	+0.17	0.5	48.74
SuperGlobal + MSPW	-	256	95.38	93.86	94.09	94.93	84.15	92.98	90.51	96.43	67.55	79.69	80.69	94.14	88.70		0.23	10.06
SENET [†] [71]	CVPR23	2048	94.86	94.41	95.16	96.08	81.41	91.64	90.53	97.86	62.26	77.33	80.30	94.00	87.99	+0.16	0.5	48.74
SENET [†] + MSPW	-	256	95.22	95.29	94.90	95.94	82.33	91.98	90.28	97.57	63.52	77.96	79.89	92.86	88.15		0.23	10.06
NetVLAD [†] [66][72]	AAAI22	1024	87.51	88.53	95.03	96.08	73.91	84.86	86.80	97.43	56.45	70.14	73.61	94.57	83.74	-0.58	0.39	26.64
NetVLAD [†] + MSPW	-	128	86.91	87.48	95.01	95.80	73.34	83.86	87.00	96.86	55.10	68.29	74.45	93.86	83.16		0.25	7.3
DUODIS [†] [65]	PAMI23	2048	82.22	85.96	92.36	95.43	63.97	83.86	76.99	99.00	38.01	54.48	55.96	89.71	76.50	+0.25	0.849	95.62
DUODIS [†] + MSPW	-	256	81.39	83.30	92.63	95.86	64.28	82.67	78.16	98.43	39.20	55.76	58.99	90.29	76.75		0.306	17.13
Whiten-MTD [†] [73]	AAAI24	512	93.52	94.26	96.34	96.51	83.01	94.27	92.34	98.43	64.13	77.54	83.21	96.43	89.16	-0.83	0.29	15.80
Whiten-MTD [†] + MSPW	-	64	92.53	92.10	95.74	95.43	82.74	93.00	92.49	97.71	63.33	74.71	84.18	94.00	88.33		0.22	5.92
Pre-trained model																		
CLIP ^{◊†} [74]	ICML21	768	60.61	68.09	80.34	94.51	46.30	64.15	70.46	96.14	18.18	29.85	49.50	83.29	63.45	+1.21	0.42	39.20
CLIP ^{◊†} + MSPW	-	96	60.67	67.36	84.31	93.65	46.65	62.20	76.74	95.71	18.89	29.29	57.73	82.71	64.66		0.24	10.19
BLIP ^{◊†} [75]	ICML22	768	44.48	52.55	80.04	93.29	36.03	53.90	66.27	97.14	13.96	20.36	42.89	73.14	56.17	+0.16	0.32	21.11
BLIP ^{◊†} + MSPW	-	96	43.83	50.33	81.86	92.64	35.80	52.75	68.07	95.86	14.08	20.80	45.48	74.43	56.33		0.23	6.61
IBOT [†] [76]	ICLR22	768	46.69	55.68	76.76	91.43	34.16	50.95	60.36	94.14	10.72	18.79	32.76	57.71	52.51	+0.69	0.33	21.11
IBOT [†] + MSPW	-	96	45.22	52.26	78.17	91.14	34.42	49.10	63.00	93.86	12.39	20.56	36.72	61.57	53.20		0.23	6.61
RADIOv2 ^{◊†} [77]	CVPR24	3072	76.98	83.33	95.14	97.29	61.45	75.38	90.16	98.14	35.63	43.43	77.04	92.57	77.21	+0.2	0.69	70.84
RADIOv2 ^{◊†} + MSPW	-	384	77.89	82.55	95.39	97.00	62.67	74.76	90.77	97.86	36.11	42.57	78.48	92.86	77.41		0.27	12.83
DINOv2-reg ^{◊†} [41]	ICLR24	1024	91.42	94.26	95.70	97.14	79.42	91.71	93.11	98.29	60.79	72.86	84.61	95.71	87.92	+0.07	0.478	50.25
DINOv2-reg ^{◊†} + MSPW	-	128	92.34	94.41	94.43	96.07	81.06	90.71	92.31	98.00	63.36	74.57	83.94	94.71	87.99		0.226	11.57

TABLE V
PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH SEVEN OTHER DIMENSIONALITY REDUCTION METHODS ON FINE-GRAINED RETRIEVAL DATASETS. BOLD NUMBERS INDICATE THE BEST RETRIEVAL PERFORMANCE AFTER FEATURE DIMENSIONALITY REDUCTION.
‡: CODE WE REPRODUCED. †: OFFICIAL CODE

Method	Dim	CUB-200-2011		Cars-196		Dogs		Time (s)	Memory (MB)
		Recall@1	Recall@5	Recall@1	Recall@5	Recall@1	Recall@5		
Original	1024	85.26	95.39	90.18	97.60	87.60	95.92	257.5	884.33
+ PWs [21]	64	82.91	94.32	86.69	96.53	86.52	95.57	8.61	885.5
+ SVD whitening‡ [22]	64	83.02	94.46	86.88	96.67	86.50	95.56	9.35	886.61
+ GWL [†] [12]	64	80.91	94.03	86.85	96.57	85.71	95.62	25.17	919.17
+ PCApw [†] [40]	64	83.36	94.41	87.09	96.79	86.64	95.57	8.38	885.5
+ UMAP [†] [81]	64	81.33	93.37	84.33	95.51	83.83	94.23	87.35	1001.72
+ MDE [†] [82]	64	74.92	90.66	79.83	93.42	82.51	93.52	41.59	884.73
+ TriMap [†] [83]	64	74.89	91.44	78.09	93.0	81.12	93.36	257.0	886.19
+ MSPW (Ours)	64	83.36	94.55	87.36	96.90	86.96	95.79	17.88	904.28

MSPW method include medical image retrieval, fine-grained retrieval, and social media image search.

2) *Limitations*: Although MSPW achieves excellent performance on several benchmark datasets, there are some limitations. First, the optimal setting of the noise-perturbation

parameter remains an open issue, which may affect the generalizability. Second, the computational cost needs to be reduced to better suit real-time applications in the online QSL stage. Third, the strategy of determining weights should become more adaptive and intelligent in the dimensional fusion stage.

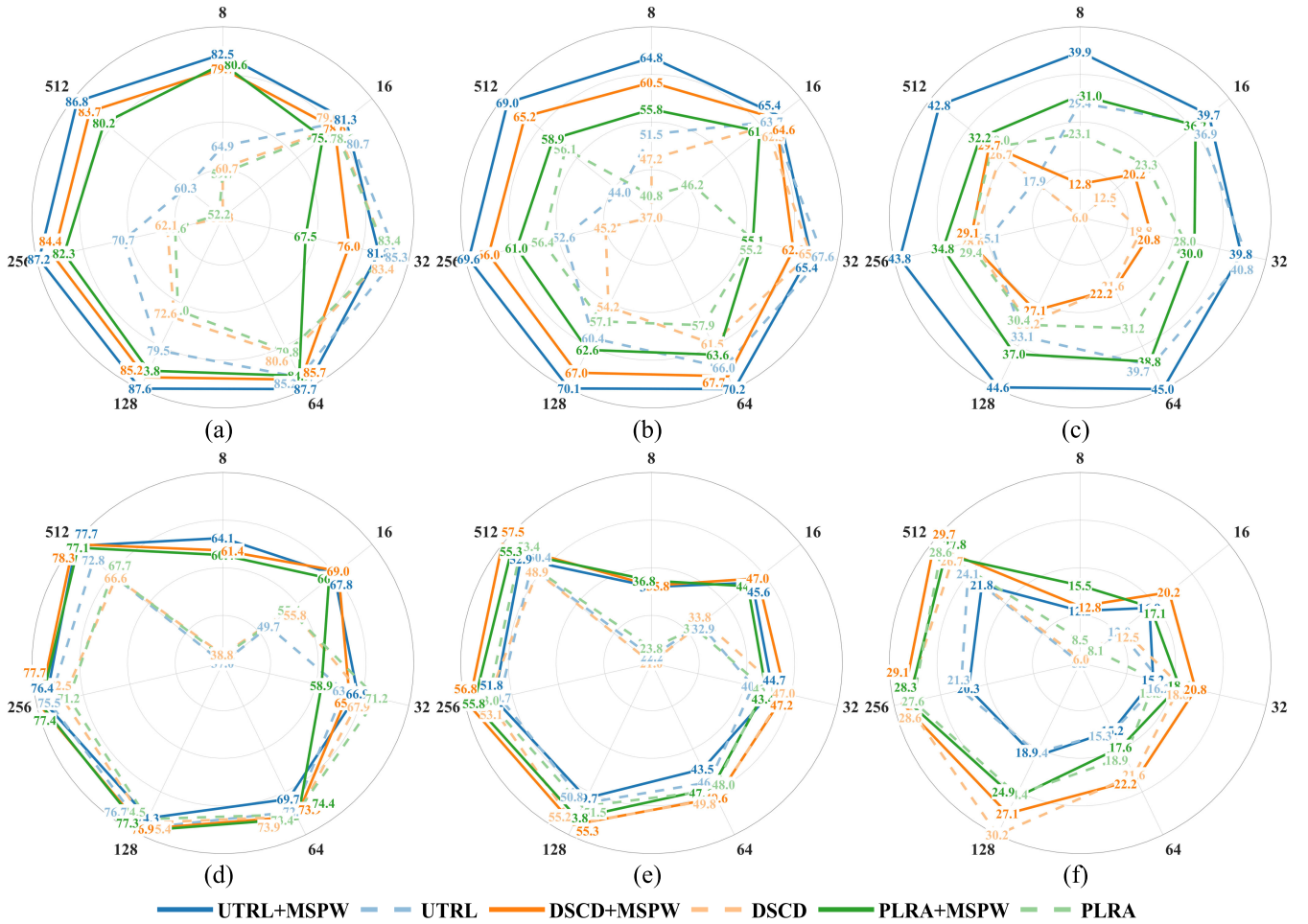


Fig. 10. Performance comparison of UTRL, DSCD, and PLRA with and without MSPW across different feature dimensions on four benchmarks, showing that adding MSPW consistently boosts mAP under the same backbone and whitening setting.

V. CONCLUSION

In this work, we propose a novel MSPW learning framework, which can effectively compress features without the need for auxiliary datasets, thereby addressing the main shortcomings of existing methods. The core idea of this method is to dynamically learn PW parameters by reconstructing database features and incorporating query features. Specifically, we first introduce an FSL method to mitigate the performance degradation problem of high-dimensional features. Second, a QSL strategy is employed to enhance the discriminability of low-dimensional features. Finally, an FF approach is proposed to generate more robust and compact representations.

The significance of our MSPW method extends beyond achieving state-of-the-art retrieval performance, and it also demonstrates remarkable flexibility and generalization. Extensive experiments have validated its effectiveness across multiple datasets and visual tasks. Furthermore, the experimental results confirm that MSPW is a plug-and-play solution that can further improve the retrieval performance of existing methods.

In future work, we plan to extend the MSPW method in a multimodal retrieval environment, using MSPW to fuse cross-modal features between different modalities to improve the effect of multimodal retrieval.

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